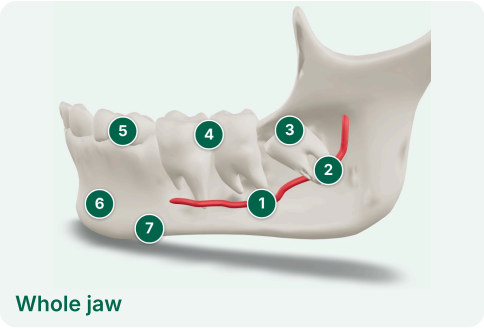
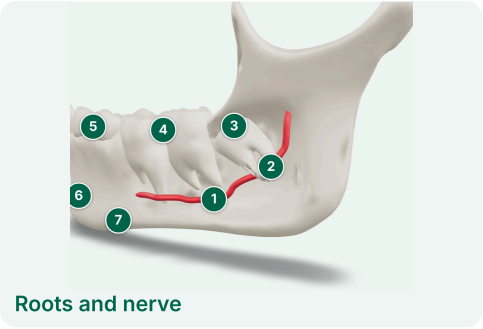
Impacted lower wisdom tooth and the nerve near it

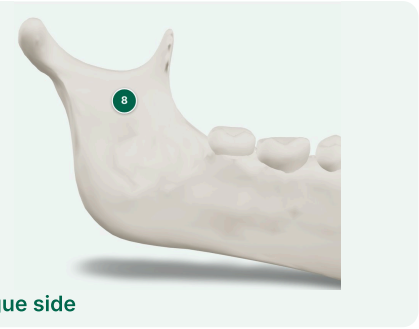
NAGOYA UNIVERSITY



Whole jaw



Roots and nerve



Tongue side

Jaw bone and teeth Nerve, artery and vein (in their canal)

1 Jaw nerve (inferior alveolar nerve)
2 Roots of the wisdom tooth
3 Wisdom tooth (third molar)
4 First and second molars
5 First and second premolars
6 Chin nerve (mental nerve)
7 Jaw bone (mandible)
8 Where the nerve goes into the bone (lingula)

What is the problem?

The wisdom tooth is the last tooth at the back. Its other name is the **third molar**. An **impacted lower wisdom tooth** is stuck under the gum and bone. It has not come through the gum (it is **unerupted**). It cannot grow in the right way.

The nerve

A nerve runs in a tube of bone inside the lower jaw. Here it is called the **jaw nerve**. Its medical name is the **inferior alveolar nerve**. Two blood vessels run with it in the same tube. They are the **inferior alveolar artery and vein**.[1]

The jaw nerve is a **branch** of the **trigeminal nerve**, the main nerve of feeling in the face. It comes from the lowest of its 3 main parts, the **mandibular division**. The nerve, artery and vein go into the jaw bone through a hole on the tongue side. This hole is the **mandibular foramen**.[1] A small spike of bone at its front edge is the **lingula**.[2]

The jaw nerve gives feeling to the lower teeth and gums on the same side.[3] Near the front of the jaw, a branch of it comes out of the bone through a small hole. This branch is the **chin nerve** (**mental nerve**). It gives feeling to the lower lip and chin.[4]

The tongue gets its feeling from a different nerve, the **lingual nerve**. It can also be hurt when a wisdom tooth is taken out.[5] Lasting damage to it is rare. It happens to about 1 to 3 in 1,000 people.[6] The tongue nerve is not shown in this model.

In this model, **the roots of the wisdom tooth wrap around the nerve and artery**.

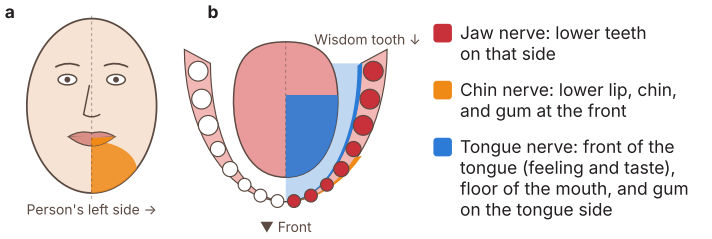
Why taking the tooth out can hurt the nerve

To take the tooth out, the roots must come out of the bone. The roots curve around the nerve. So the nerve can be pressed, stretched or cut as the roots come out. The artery and vein can also be hurt and bleed.[1]

If the jaw nerve is hurt, the chin nerve may also stop working well. This is because it is a branch of the jaw nerve.[4]

What nerve damage feels like

The lower lip, chin, teeth or gums on that side may feel numb, tingle, feel strange or hurt.[3,5]



Where each nerve gives feeling, on the side of the wisdom tooth. If a nerve is hurt, these areas can feel different. **a:** the face from the front. **b:** the lower teeth and tongue, seen from the front.[7,8] Drawing made for this study.

Short-term: the feeling comes back, most often within weeks to months. **Long-term:** the feeling does not fully come back.[3]

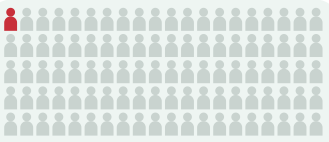
In older people, the risk is higher and the feeling is less likely to come back fully.[3,9]

How often does it happen?

Some wisdom teeth have roots close to the nerve. A review of 16 studies looked at these teeth. It found:[10]



About 7 in 100
had nerve damage after the whole tooth was taken out



About 1 in 100
had nerve damage that did not go away

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